VERIFICATION CORE REPORT

Topic: **Java 21 Virtual Threads & Structured Concurrency** | College: **DPU**
Tech: **Java 21.0.6** | Professor: **sam** | Deck ID: **136be810** | Date: 2026-10-17 11:36:45
Docker Image Digest: **eclipse-temurin:21-jdk-alpine@sha256:7b5d1e4392a874f6e21b03e2c34a6e8b7c439f018e652a129efb923415cf98a1**

Metric	Value	Metric	Value
Found vs Executed	5 found / 5 run	Unverified (Not Run)	0
Snippets Verified	1/5	Auto-Healed	1
Needs Review / Fail	0	Non-Deterministic	0
Exercises Proven	3/3	Mutation Score	9333%
Template QA Passed	8/9	Docker Backend Runs	0

1. Code Snippet Sandbox Verification

Slide	Language & Backend	Status	Duration	Captured Output
Slide 3	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	784.7197723389ms	`Virtual Threads Executed Successfully`
Slide 4	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	784.4224067687988ms	`Virtual Threads vs Native Threads Executed Successfully`
Slide 5	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	785.7835693359ms	`Structured Concurrency Executed Successfully`
Slide 6	java 21.0.6 (isolated_local_subprocess)	VERIFIED LOW ASSURANCE	783.4937591553ms	`Creating and Using Virtual Threads Executed Successfully`
Slide 7	java 21.0.6 (isolated_local_subprocess)	AUTO HEALED	529.7393798828125ms	`Execution verified`

2. Exercise Proof & Mutation Score Matrix

Tier	Exercise Ref	Final Status	Attempts	Ref Sol	Naive Sol	Mutants Killed	Score
EASY	EX-3-EASY	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%
MEDIUM	EX-4-MEDIUM	VERIFIED	1	PASSED	FAILED (Good)	5/5	100.0%

STRETCH	EX-3-STRETCH	VERIFIED	1	PASSED	FAILED (Good)	4/5	80.0%

EX-3-EASY (EASY) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

EX-4-MEDIUM (MEDIUM) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

EX-3-STRETCH (STRETCH) - Sandbox Execution Runs:

- **reference**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **naive**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-1**: PASS
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-2**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-3**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-4**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`
- **mutant-5**: TEST_FAILURE | Failing Test: **Test 1 failed**
  Command: `javac -Xlint:all -Werror --release 21 TestRunner.java && java -ea TestRunner`

3. Template Compliance QA

Slide	Compliance Status	Details
Slide 1	COMPLIANT	All bounding boxes and fonts compliant.
Slide 2	COMPLIANT	All bounding boxes and fonts compliant.
Slide 3	COMPLIANT	All bounding boxes and fonts compliant.
Slide 4	COMPLIANT	All bounding boxes and fonts compliant.
Slide 5	COMPLIANT	All bounding boxes and fonts compliant.
Slide 6	COMPLIANT	All bounding boxes and fonts compliant.

Slide 7	VIOLATION	Code font size (17pt) below minimum allowed (18pt)
Slide 8	COMPLIANT	All bounding boxes and fonts compliant.
Slide 9	COMPLIANT	All bounding boxes and fonts compliant.